

Shared Attribute Supervision for Respiratory Sound Learning Across Native Tasks

Abstract—Respiratory-sound datasets describe related acoustic events but differ in annotation units, label granularity, and coverage, complicating the joint use of their supervision. We present an annotation-aware framework that shares respiratory-sound attribute supervision while retaining dataset-specific classification targets. A shared BEATs encoder is jointly trained on ICBHI respiratory cycles and SPRSound events to predict abnormality, crackle, and wheeze using annotation-supported targets. A two-level readout uses abnormality for SPRSound’s binary task and combines it with crackle/wheeze predictions for ICBHI’s four-class task. HF Lung provides a separate auxiliary-supervision study and attribute evaluation, while KAUH is reserved for external patient-level assessment. Across three exploratory runs with ICBHI-test-based checkpoint selection, the core model achieves an ICBHI Score of $61.17 \pm 0.31\%$ and a SPRSound official Score of $90.70 \pm 0.34\%$. A separate single-run comparison with added HF supervision shows higher SPRSound Score but lower ICBHI abnormal-subclass recall. These results demonstrate joint native-task prediction and characterize the behavior of shared attributes under additional supervision and external evaluation.

Index Terms—respiratory sounds, shared supervision, heterogeneous annotations, acoustic attributes, hierarchical decoding

I. INTRODUCTION

Respiratory-sound datasets provide complementary observations of related acoustic events, but their annotations answer different questions. ICBHI assigns Normal, Crackle, Wheeze, or Both to complete respiratory cycles. SPRSound annotates individual sound events and defines both coarse Normal/Adventitious classification and finer event categories [1], [2]. HF Lung provides timed breath-phase and adventitious-sound annotations [3], whereas KAUH provides recording-level sound labels and three filtered versions of each patient’s recording [4]. Differences in annotation units, category granularity, and coverage therefore accompany acoustic variation across sources. Using these datasets together requires deciding which supervision can be shared and which distinctions each task must retain.

A shared acoustic representation offers a way to connect these tasks. Crackle and Wheeze annotations can supervise the same attributes across sources, while an abnormality label supplies a coarser distinction. However, these training signals carry different amounts of information. An abnormality label alone does not specify a fine sound attribute, and merging fine categories can remove distinctions required by another task. The central question is whether compatible attribute supervision can preserve useful acoustic information across these annotation conditions and support the original classification targets through a common model.

Several lines of work provide the foundations for this question. SPRSound includes an ICBHI/SPRSound data-fusion benchmark [2], and LungMix unifies respiratory labels into a common four-class space for single-source generalization [5]. OPERA learns shared respiratory representations and evaluates them with separate downstream predictors [6]. In heterogeneous supervised learning, universal taxonomies map shared predictions back to dataset-specific labels [7], and DCASE 2024 combines audio datasets with label mapping and missing-target masking [8]. Respiratory attribute classification also has close precedents: PC-MCL uses explicit multi-label targets and deterministic four-class conversion [9], while OCAD develops event factors and structured compositional decoding [10]. Our focus is the specific relationship between shared attribute supervision, annotation conditions, and native respiratory-task readouts.

We organize the study around all four datasets. ICBHI and SPRSound form the core joint-training pair. Shared abnormality and C/W predictions return ICBHI cycle-four-class and SPRSound event-binary outputs through a two-level decision scheme. HF Lung tests an additional source of attribute supervision in a separate condition and provides source-test attribute evaluation. KAUH remains external to training, threshold fitting, and model selection. This organization makes it possible to examine both how annotations enter learning and how a fixed shared model behaves on other sources. A binary training task can use available fine labels, while interval and recording annotations provide different tests of the learned attributes.

The draft is organized around three contributions:

- 1) An annotation-aware joint learning framework that shares respiratory attributes across cycle and event annotations and returns their native classification targets through a two-level decision scheme.
- 2) A characterization of shared attributes under differing annotation coverage, using HF supervision tests and attribute evaluation to examine what another source’s labels contribute.
- 3) An analysis of attribute information and task decoding through core and external evaluations, with supervision-matched comparisons designed to separate their effects on task performance and class-specific trade-offs.

Draft note. Core, HF, and KAUH results below are existing benchmark or exploratory evaluations. The matched annotation and classification controls in Sections IV-G and IV-H remain pending.

Core native annotation	a	c	w
Normal (ICBHI / SPRSound)	0	0	0
Crackle / Coarse or Fine Crackle	1	1	0
Wheeze (ICBHI / SPRSound)	1	0	1
Both / Wheeze+Crackle	1	1	1
Rhonchi or Stridor (SPRSound)	1	–	–

Fig. 1. Core shared supervision. A dash denotes an unavailable target. HF’s separate annotation-derived supervision convention is defined in Section III-D; KAUH contributes evaluation labels only.

II. FOUR DATASETS AND SUPERVISION ROLES

A. Study Organization

Table I summarizes the four datasets. ICBHI and SPRSound train the core model and supply its native-task benchmarks. HF Lung has two roles: testing additional attribute supervision using its source-training portion, and evaluating predictions on its source-test portion. KAUH supplies an external test with no training or calibration. The HF-assisted condition is separate from the two-source core model.

B. ICBHI and SPRSound: Core Training and Native Tasks

ICBHI provides cycles labeled Normal, Crackle, Wheeze, or Both [1]; Both denotes C/W co-presence within the cycle. Its official recording split has 4,142 training and 2,756 test cycles. The test contains 1,579 Normal, 649 Crackle, 385 Wheeze, and 143 Both cycles. This official split is not strictly patient-held-out. SPRSound provides fine event annotations [2]; our native endpoint is BioCAS2022 Task1-1 on the official inter-subject test, with 1,429 events (1,040 Normal and 389 Adventitious). Its intra-subject test and seven-class Task1-2 are distinct endpoints.

Let a , c , and w denote abnormality, Crackle, and Wheeze. ICBHI labels determine all three. Compatible SPRSound labels supply the same targets, merging Coarse and Fine Crackle at the attribute level. Rhonchi and Stridor supervise abnormality but leave C/W unavailable. Figure 1 shows this core mapping; unavailable targets do not enter the corresponding classification loss.

The attributes express the two selected core tasks. They do not retain every source distinction, such as Coarse versus Fine Crackle, so preserving these endpoints does not imply recovery of SPRSound’s entire seven-class taxonomy.

C. HF Lung: Supervision Test and Attribute Evaluation

HF_Lung_V1 contains 9,765 recordings of 15 s, with 7,809 source-training and 1,956 source-test recordings [3]. Its timed annotations include inhalation (I), exhalation (E), discontinuous adventitious sounds (D), Wheeze, Rhonchi, and Stridor. The D category consists of crackles; the native benchmark also groups Wheeze, Rhonchi, and Stridor into continuous adventitious sounds. Our shared heads evaluate D/Crackle and Wheeze rather than reproduce the full native I/E/CAS/DAS temporal task.

The existing HF supervision test divides source-training data by recording-date proxies into 5,322 fitting and 2,487

monitoring recordings. Date proxies are not patient identifiers. Only the fitting portion contributes auxiliary updates; monitoring does not choose checkpoints or thresholds. Each recording produces three continuous 5-s windows. Source-test data remain reserved for selected-checkpoint evaluation. The historical auxiliary target rule and its negative-label assumption are specified in Section III-D.

D. KAUH: External Patient-Level Assessment

KAUH contains one base recording per subject, exported through Bell (B), Diaphragm (D), and Extended (E) filters: 336 files from 112 subjects [4]. B/D/E are correlated versions of the same recording. We retain the patient identifier for aggregation and use the sound label, not the disease diagnosis, to define compatible evaluation targets.

The mapping is N→Normal; C or I C→Crackle; E W or I E W→Wheeze; and I C E W→Both. Crep, Bronchial, and I C B remain outside the current mapping. The scored subset contains 258 files from 86 patients: 35 Normal, 8 Crackle, 41 Wheeze, and 2 Both. All 112 patients’ files are retained in the external prediction set, while the reported metrics use this supported subset. The result is a compatible-label assessment, not native nine-class reproduction.

E. Audio Preparation

All model inputs are mono at 16 kHz. ICBHI cycles, SPRSound events, and KAUH files use the existing 5-s input construction: front truncation for long units and repetition for short ones. The inherited pipeline applies a 1/16-s recording-boundary fade and an end fade to repeated inputs. HF instead uses [0,5], [5,10], and [10,15] s without creating new temporal labels. Predictions remain associated with their original cycle, event, recording, or patient; long-unit truncation does not establish full event coverage.

III. METHOD

A. Shared Acoustic Representation and Attributes

Figure 2 separates the acoustic input path from the annotation path. We initialize the encoder with BEATs iter3+ AS2M [11] and fine-tune it jointly on the two datasets. Its 768-dimensional token features are mean-pooled and linearly projected to $h_i \in \mathbb{R}^{256}$. Three parallel linear predictors produce

$$\begin{aligned} p_{ia} &= \text{softmax}(W_a h_i + b_a)_{\text{Abnormal}}, \\ p_{ic} &= \sigma(W_c h_i + b_c), \quad p_{iw} = \sigma(W_w h_i + b_w). \end{aligned} \quad (1)$$

The abnormality predictor has two logits; each attribute predictor has one. All three update the shared representation. The two-level structure is used for task decisions: the C/W probabilities are marginal predictions, with no explicit parent-child probability-consistency constraint.

TABLE I
FOUR DATASETS AND THEIR ROLES FOR THE SHARED MODEL. AUXILIARY HF SUPERVISION IS USED ONLY IN ITS SEPARATE SUPERVISION-TEST CONDITION. HF SOURCE-TEST AND KAUH DO NOT UPDATE THE MODEL OR DETERMINE ITS SELECTION OR THRESHOLDS.

Dataset	Annotation and unit	Learning role	Evaluation in this study
ICBHI	Respiratory cycles; Normal, Crackle, Wheeze, Both	Core joint training; abnormality and C/W targets	Official cycle four-class task
SPRSound	Respiratory events; seven source event categories	Core joint training; abnormality and supported C/W targets	Official inter-subject event binary task
HF Lung	15-s recordings; timed I/E and D/W/R/S annotations	Separate auxiliary attribute-supervision test, source-train only	Source-test positive-interval coverage and recording attribute presence
KAUH	Recording sound labels; B/D/E versions of each patient recording	None; external evaluation only	Patient-level binary and compatible four-class readouts

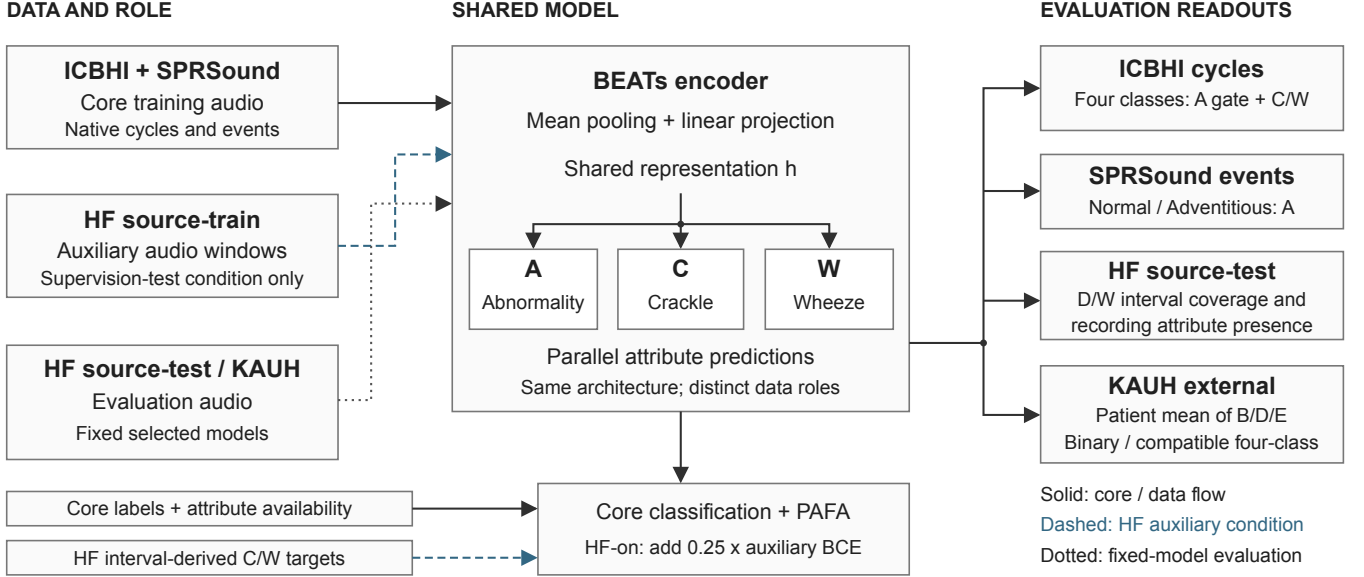


Fig. 2. Four-dataset study using shared predictions. ICBHI and SPRSound provide core training audio and labels. HF source-train enters only the auxiliary-supervision condition; HF source-test and KAUH are evaluated with fixed selected models. Three parallel predictors support dataset-specific readouts. Labels enter losses, and the two-level decision structure does not make the predictors a cascade.

B. Learning from Available Supervision

For sample i and node $k \in \{a, c, w\}$, $m_{ik} = 1$ when its source annotation supports target y_{ik} . For batch B , let $B_k = \{i \in B : m_{ik} = 1\}$ and $K_B = \{k : |B_k| > 0\}$. The implemented classification objective averages over available samples within each node:

$$\mathcal{L}_{\text{cls}}(B) = \frac{1}{|K_B|} \sum_{k \in K_B} \frac{1}{|B_k|} \sum_{i \in B_k} \ell_k(p_{ik}, y_{ik}), \quad (2)$$

where ℓ_a is two-class cross-entropy and ℓ_c, ℓ_w are binary cross-entropy. Dataset-homogeneous batches contribute equally many updates per epoch. Thus supported labels from either source train the shared attributes without requiring every sample to supervise every node.

We retain PAFA’s patient cohesion-separation and global patient alignment objectives [12]:

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + 50\mathcal{L}_{\text{PCSL}} + 0.0005\mathcal{L}_{\text{GPAL}}. \quad (3)$$

An auxiliary attention-based projection supplies their features, with patient identifiers qualified by dataset. This auxiliary

projection is used during training and does not enter the native classification readouts.

C. Two-Level Native Task Decisions

The Normal/Abnormal argmax defines $\hat{a}$; its ties select Normal. Attribute decisions are $\hat{c} = \mathbb{I}[p_c \geq \tau_c]$ and $\hat{w} = \mathbb{I}[p_w \geq \tau_w]$. SPRSound directly uses $\hat{a}$ for Normal/Adventitious classification. ICBHI assigns Normal if $\hat{a} = 0$. For an abnormal decision, two positive attributes yield Both and one positive attribute yields its corresponding class. If neither is positive, the larger $p_k - \tau_k$ margin selects Crackle or Wheeze, with ties assigned to Crackle.

For the reported runs, the shared C/W thresholds independently maximize the equal-dataset mean of eligible-node binary F1 on internal validation data. Candidates are the observed probabilities together with 0 and 1; exact ties select the larger threshold. This readout returns task labels from the same attribute scores. Its contribution can therefore be studied separately from representation learning by fixing the scores and changing only the decision rule.

D. HF Auxiliary-Supervision Condition

The existing HF experiment retains the core architecture and adds

$$\mathcal{L}_{\text{HF-on}} = \mathcal{L}_{\text{core}} + 0.25\mathcal{L}_{\text{HF}}, \quad (4)$$

where $\mathcal{L}_{\text{core}}$ includes the classification and PAFA terms above. One batch of 32 HF windows accompanies each core update. For a window that overlaps any D, Wheeze, Rhonchi, or Stridor annotation, both C/W targets are eligible. A target is one if its corresponding D or Wheeze annotation overlaps the window and zero otherwise. Node-wise BCE values are averaged. Pure gaps, empty annotations, and phase-only windows are excluded; HF does not supervise the abnormality head.

This historical implementation uses annotation-derived negative targets on eligible abnormal windows, including Rhonchi/Stridor-only windows. It is not a positive-only objective. Because the source annotations may omit uncertain events [3], these negatives embody an annotation-completeness assumption. HF is excluded from threshold fitting, checkpoint selection, and early stopping; the HF source-test partition does not enter training.

E. HF and KAUH Evaluation Readouts

HF interval coverage counts a labeled D or Wheeze interval as recognized when any overlapping 5-s window exceeds the fixed corresponding attribute threshold. It measures coverage of annotated positive intervals, not precise onset/offset localization. Recording-level presence uses the maximum score across the three windows. Its evaluation pool contains recordings with at least one D/W/R/S annotation; an absent target in this pool is treated as a derived negative. Thus these curves assess discrimination among annotated abnormal recordings, not detection of Normal versus Abnormal on all HF recordings.

For KAUH, probabilities are averaged across a patient’s B/D/E files before decoding. The abnormality decision gives the binary output, while the same two-level rule gives four compatible classes. Neither external dataset supplies new calibration: each selected model uses its core-validation thresholds. Recordings are grouped by patient before reporting the primary KAUH metrics.

IV. EVALUATION

The four-dataset evaluation follows their distinct roles: native task performance on the ICBHI/SPRSound training pair, HF attribute evaluation and supervision testing, and KAUH external assessment. Existing core and external results are reported below. Further matched comparisons isolate annotation information and the final classification design.

A. Protocol and Comparison Sources

The joint model uses full BEATs fine-tuning, batch size 32, Adam with learning rate 5×10^{-5} and weight decay 10^{-6} , a 50-epoch cosine schedule with minimum learning-rate ratio 0.001, and parameter EMA 0.5. Each epoch has 326 updates, divided equally between the sources. SpecAugment, mean-variance normalization, and HF auxiliary supervision are disabled.

TABLE II

EXISTING NATIVE-TASK RESULTS, IN PERCENT. SINGLE-DATASET ROWS USE SEPARATELY TRAINED MODELS FOR THE TWO COLUMNS; THE JOINT ROW USES ONE MODEL PER SEED. FROZEN-ENCODER VALUES ARE REPORTED SINGLE-RUN REFERENCES. RECIPES AND SELECTION PROCEDURES DIFFER, SO THE ROWS ARE NOT MATCHED ABLATIONS.

Model / training setting	ICBHI Score	SPRSound inter Official Score
<i>Single-dataset frozen encoders: seed 42</i>		
AST	51.22	89.99
BEATs	51.37	92.00
PANNs	51.97	91.31
OPERA-CT	38.61	79.45
HeAR	48.98	88.12
<i>ICBHI source checkpoint references</i>		
PAFA [12]	64.14	–
SG-SCL [13]	60.98	–
Patch-Mix [14]	62.17	–
<i>Joint full fine-tuning: three runs</i>		
Shared attributes	61.17 ± 0.31	90.70 ± 0.34

Internal ICBHI validation uses the first fold of a shuffled five-fold stratified patient-group split, controlled by the run seed. SPRSound’s 6,656 official training events are divided into 5,219 fitting and 1,437 validation events, with 194 and 49 patient groups. These data set attribute thresholds. In the existing runs, checkpoint selection and patience-10 early stopping instead use ICBHI official-test Score. The formal seeds 0, 1, and fresh 42 select epochs 15, 11, and 17; SPRSound official inter is evaluated once per selected checkpoint. The resulting mean and sample standard deviation describe a test-selected benchmark. New matched comparisons will use a common validation-based selection protocol, with its split, criterion, and training budget fixed before execution.

For ICBHI, Sp is Normal recall and Se is the fraction of abnormal cycles assigned to their correct abnormal subclass. Its Score is $(Sp + Se)/2$; confusing Crackle with Wheeze therefore reduces Se . For SPRSound Task1-1, Se and Sp are Adventitious and Normal recall. Its official Score is $(AS + HS)/2$, where $AS = (Se + Sp)/2$ and $HS = 2SeSp/(Se + Sp)$.

The frozen-encoder references in Table II come from the single-dataset baseline report. They use 16-kHz audio, 2-s windows with 1-s hop and tail coverage, zero-padding, a 256-dimensional projector and native head, batch size 8, 50 epochs, and seed 42. ICBHI uses official-test Score selection; SPRSound intra/inter runs are selected separately. The source checkpoint rows are local evaluations of ICBHI-specialized models, not the corresponding published multi-run means. These references provide context for the native tasks, with training and input differences made explicit.

B. Native-Task Performance with a Shared Model

The joint model obtains ICBHI Score $61.17 \pm 0.31\%$ and SPRSound official Score $90.70 \pm 0.34\%$. Its ICBHI Sp and Se are $74.98 \pm 5.69\%$ and $47.35 \pm 5.35\%$. Class recall is $60.20 \pm 8.48\%$ for Crackle, $29.61 \pm 2.08\%$ for Wheeze, and

TABLE III

FIXED-CHECKPOINT EVALUATION OF THE THREE CORE MODELS ON HF AND KAUH. ALL VALUES, INCLUDING AUROC, ARE PERCENTAGES (MEAN $\pm$ SAMPLE SD). HF USES ITS SOURCE-TEST SET; KAUH USES 86 COMPATIBLE PATIENTS. NO EXTERNAL THRESHOLD TUNING OR MODEL RESELECTION IS PERFORMED.

Dataset / target	Metric	Core model
HF D/Crackle	Interval recall	58.28 $\pm$ 12.63
	Recording AUROC	67.82 $\pm$ 4.42
HF Wheeze	Interval recall	85.43 $\pm$ 1.07
	Recording AUROC	86.36 $\pm$ 1.38
KAUH binary	Balanced accuracy	72.17 $\pm$ 1.64
KAUH compatible four-class	Average score	47.00 $\pm$ 8.30
	Macro-F1	33.83 $\pm$ 3.17
	UAR	50.24 $\pm$ 6.10

36.83 $\pm$ 11.37% for Both. SPRSound Normal and Adventitious recall are 84.29 $\pm$ 1.36% and 97.60 $\pm$ 1.21%, respectively. Thus the shared model supplies both requested outputs, with uneven recognition of the fine ICBHI categories. The small ICBHI Score variation does not imply equally stable class behavior.

The frozen BEATs references obtain 51.37% on ICBHI and 92.00% on SPRSound, whereas the single-source PAFA checkpoint reaches 64.14% on ICBHI. These different reference points do not establish a sharing gain or cost: they also differ in fine-tuning, inputs, losses, and selection. A matched joint-versus-single-source comparison is needed to assess performance retention and the value of adding another source’s supervision.

C. HF Attribute Evaluation with Fixed Core Models

The three selected core models are evaluated on all 1,956 HF source-test recordings, using 5,868 continuous 5-s windows and their fixed core-validation thresholds. The positive-interval denominators are 1,812 D and 1,430 Wheeze intervals. Recording-presence curves use 957 recordings containing at least one adventitious annotation: 368 have D and 405 have Wheeze. Other recordings are excluded from this presence comparison, and it uses the derived-negative convention in Section III-E.

Table III shows higher interval coverage and recording-level discrimination for Wheeze than for D/Crackle in this setting. Interval recall counts coverage of labeled positive events and does not measure false positives in unannotated regions. The results therefore characterize transfer of the two shared attributes rather than complete HF temporal-event detection or Normal/Abnormal classification.

D. HF Supervision Test and Cross-Dataset Effects

A historical seed-42 experiment adds the HF auxiliary loss described in Section III-D. It uses the same core recipe and 326 core updates per epoch, with one additional HF batch per update. HF monitoring is excluded from threshold fitting, checkpoint selection, and early stopping; ICBHI official-test Score selects the checkpoint. The source-test partition is accessed after selection. This is an exploratory supervision comparison. Its earlier HF-off reference is separate from

TABLE IV

HISTORICAL HF SUPERVISION TEST: SEPARATE SEED-42 HF-OFF AND HF-ON RUNS, IN PERCENT. HF-OFF IS THE EARLIER CORE REFERENCE, NOT THE THREE-RUN MEAN. BOTH USE ICBHI-TEST-BASED SELECTION; THE HF-ON RULE INCLUDES ANNOTATION-DERIVED NEGATIVE TARGETS ON ELIGIBLE ABNORMAL WINDOWS.

Dataset / metric	HF-off	HF-on
ICBHI Score	60.05	59.71
ICBHI abnormal-subclass recall	45.37	39.25
SPRSound official Score	89.20	91.81
HF D interval recall	52.65	93.10
HF D recording AUROC	50.81	69.80
HF Wheeze interval recall	93.78	76.92
HF Wheeze recording AUROC	85.50	89.83
KAUH binary balanced accuracy	74.76	70.14
KAUH four-class average score	64.96	67.20
KAUH four-class Macro-F1	51.47	36.65
KAUH four-class UAR	62.33	38.39

the formal three-run core aggregate and is kept separate in Table IV.

HF-on increases D interval recall and the two recording AUROCs, but Wheeze interval recall decreases. SPRSound Score rises while ICBHI abnormal-subclass recall falls. These outcomes indicate that the added supervision changes attribute and decision behavior differently across tasks. They do not establish universal improvement, and the annotation-derived negatives prevent interpreting this historical run as a test of a strictly positive-only objective.

E. KAUH External Patient-Level Evaluation

KAUH is unused by both the core and HF-assisted models until external evaluation. All B/D/E probabilities are pooled within patient before applying the fixed readout. Binary balanced accuracy is the mean of Normal and Abnormal recall. The compatible four-class average score is the mean of Normal recall and pooled correct abnormal-subclass recall; it is distinct from four-class UAR, which weights the four classes equally.

For the three core models, binary balanced accuracy is 72.17 $\pm$ 1.64%, while the compatible four-class average score is 47.00 $\pm$ 8.30% (Table III). These are different endpoints, with only two Both patients in the scored subset. They cannot be subtracted to quantify a single-task loss. In the historical HF-on comparison, the four-class average score rises slightly, but Macro-F1 and UAR decrease and binary balanced accuracy falls (Table IV). This illustrates why the external evaluation reports several class-sensitive measures alongside an aggregate score.

F. Task-Specific HF and KAUH Reference Results

The supplied frozen-encoder baseline report also covers HF and KAUH, using native tasks broader than the shared readouts. HF uses I/E/CAS/DAS temporal labels, with five date-based training folds and a test-probability ensemble; fold validation mean segment F1 selects each model. KAUH uses recording-level nine-class prediction and a 78/34-patient train/validation split, with validation-based selection. These

TABLE V

REPORTED FROZEN-ENCODER REFERENCES FOR HF AND KAUH, IN PERCENT. THE HF TEMPORAL TASK AND KAUH NINE-CLASS TASK DIFFER FROM THE SHARED-MODEL READOUTS IN TABLE III; THE VALUES ARE CONTEXTUAL, NOT PAIRED PERFORMANCE COMPARISONS.

Encoder	HF I/E/CAS/DAS		KAUH raw9 Reported Score
	Segment F1	Event F1	
AST	55.26	1.77	63.75
BEATs	55.51	2.29	66.81
PANNs	50.47	1.45	63.03
OPERA-CT	43.79	1.13	43.84
HeAR	51.64	1.37	47.15

are separately trained reference systems, not additional training data for the shared model. Their one documented training seed is 42; HF’s five folds are not five independent seeds.

HF values are the report’s mean segment and event F1 measures. The reported KAUH Score is $(AS + HS)/2$, whereas our compatible-label average score uses $(Sp + Se)/2$. Differences in task, aggregation, selection, and training prevent a direct comparison with the fixed shared-model evaluations. Retaining these references makes the native task definitions visible without changing the scope of the shared predictions.

G. Attribute Reuse under Annotation Changes

Beyond the existing HF supervision test, a matched annotation study will keep joint audio and architecture fixed while withholding SPRSound C/W training targets and retaining its native abnormality supervision. Compatible C/W labels remain available for evaluation. Source-only learning and the complete-label joint model provide reference conditions. The target source still participates in coarse-task training.

Retained loss weights must stay fixed: deleting C/W from the active-node average in (2) would otherwise also change the relative weight of abnormality supervision. These comparisons require a common validation-based selection rule and predeclared attribute metrics.

Draft note. These matched annotation results remain pending. The historical HF-on experiment does not substitute for this different supervision question.

H. Separating Classification and Supervision Effects

Two comparisons address the two-level design. First, a shared encoder with native ICBHI-four-class and SPRSound-binary heads should have access to the same fine attribute annotations, for example through matched auxiliary attribute supervision. Comparing this model with the shared-attribute model assesses the classification interface at matched information content. A binary-only SPRSound head without attribute supervision would change both the available information and the classification design.

Second, fixing the shared model’s attribute scores allows a direct readout comparison. C/W-only four-class conversion maps two negative attributes to Normal, two positive attributes to Both, and a single positive attribute to its class. Comparing this rule with the abnormality-gated rule isolates changes

TABLE VI

ABLATIONS REQUIRED BY THE STORYLINE. RESULTS ARE PENDING; THE EXISTING JOINT BENCHMARK DOES NOT FILL THESE COMPARISON ROWS.

Comparison	Question answered
Joint vs. each single source	Value of additional source supervision
Full vs. coarse SPR supervision	Value and reuse of fine attributes
Shared attributes vs. native heads, same labels	Effect of classification interface
Two-level vs. C/W-only readout, fixed scores	Effect of the final decision rule

in the final decisions. Both rules require validation-based threshold selection. Since the encoder and scores stay fixed, this comparison cannot be interpreted as a change in learned acoustic features.

Draft note. The comparisons in Table VI specify questions and controls, not completed findings. Their results will determine whether two-level decoding improves a task, changes its operating point, or adds little beyond the shared supervision.

V. CONCLUSION

We study shared respiratory-sound attributes across four datasets with different annotation and evaluation roles. ICBHI and SPRSound train the core model and retain their native cycle-four-class and event-binary outputs. HF Lung provides a separate supervision test and attribute evaluation; KAUH tests external transfer through patient-level compatible readouts. Existing results demonstrate the joint core capability and characterize attribute- and task-dependent behavior beyond the training pair. The historical HF supervision condition improves some measures while reducing others, emphasizing the role of annotation conventions and decision rules. Matched annotation and classification controls remain necessary to establish how much of these changes is attributable to shared information or two-level decoding.

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